Modern Oracle Database & AI Services

The Core of Enterprise Architecture in the AI Era

SE 4458: Software Architecture & Design

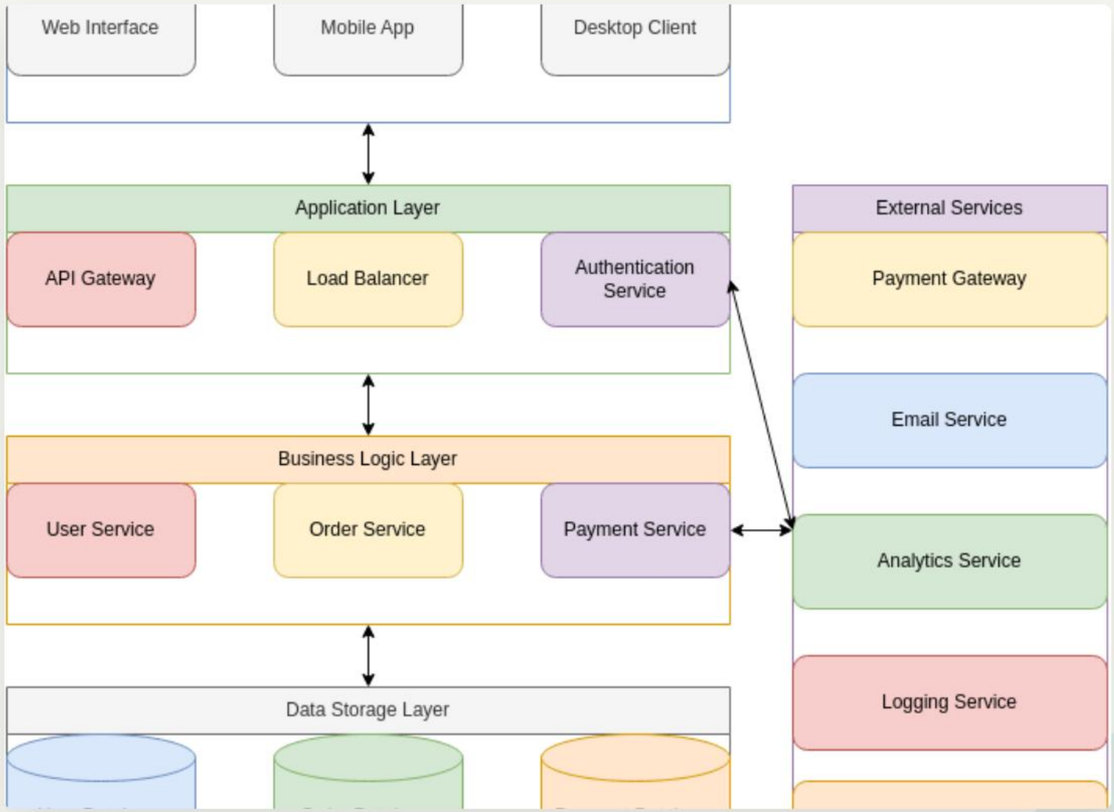
Large Scale Systems Assignment

THE "DATA CHAOS" CHALLENGE

Architectural Fragmentation

In modern microservices, developers often use specialized databases for every data type (Relational, JSON, Vector, Graph).

- High Operational Complexity
- Security Vulnerabilities in Data Silos
- Latency in Synchronizing Engines



THE SOLUTION: CONVERGED DATABASE

Multi-Model

One engine for JSON, Spatial, Graph, XML, and Relational data types.

Unified Security

Apply enterprise-grade policies across all data models simultaneously.

Synergy

Run SQL queries across JSON and Graph data in a single transaction.

Oracle acts as the "Smartphone" of databases—integrating all tools into one powerful platform.

ORACLE AUTONOMOUS DATABASE

Self-Driving Capabilities

Leveraging Machine Learning to eliminate manual administration tasks (DBA operations).

- **Self-Driving:** Auto-tuning and patching
- **Self-Securing:** Automatic encryption/threat detection
- **Self-Repairing:** 99.995% uptime guarantee

JSON-RELATIONAL DUALITY

Dev View: JSON

Work with rich JSON documents. Easy to integrate with Frontend APIs and JavaScript frameworks.

DBA View: SQL

Data is stored in ACID-compliant tables. Enables complex joins and reporting across all documents.

The Bridge

Zero data duplication. Changes in JSON are instantly reflected in Relational views and vice versa.

INNOVATION: AI VECTOR SEARCH

23ai

The Generative AI Database

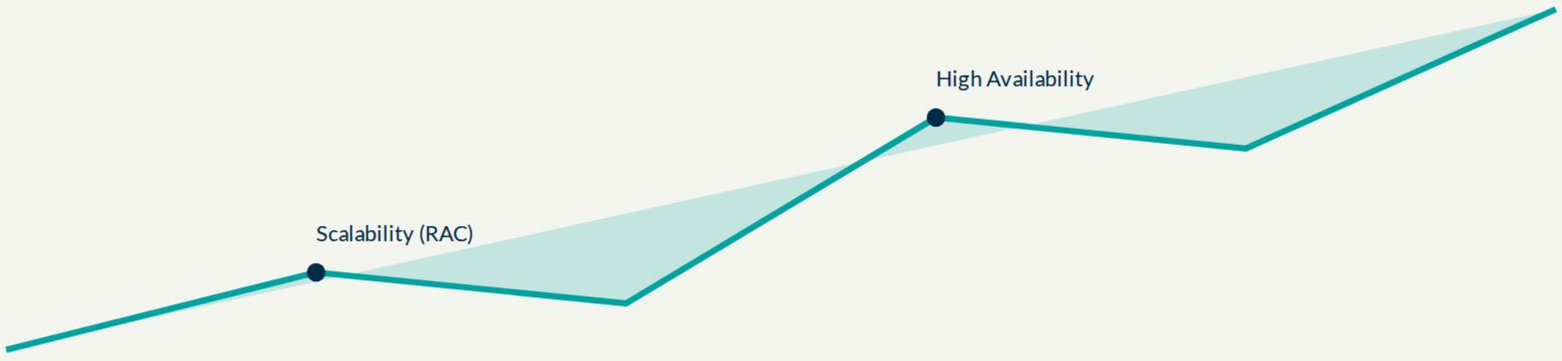
Oracle 23ai integrates Vector Embeddings into the core SQL engine.

RAG (Retrieval Augmented Generation):

Search data by *meaning* rather than keywords. Store high-dimensional vectors directly alongside business data to ground LLM responses.

```
SELECT book_id FROM library
ORDER BY VECTOR_DISTANCE(vec_col, :query_vec)
FETCH FIRST 5 ROWS ONLY;
```

SCALABILITY & AVAILABILITY



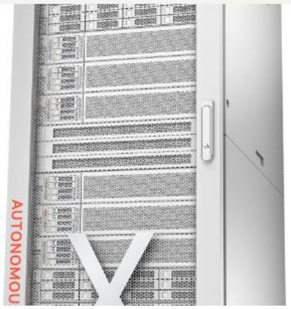
Oracle RAC: Horizontal scalability through a shared-everything cluster architecture.

Data Guard: Disaster recovery and data protection for mission-critical apps.

INFRASTRUCTURE & ECOSYSTEM

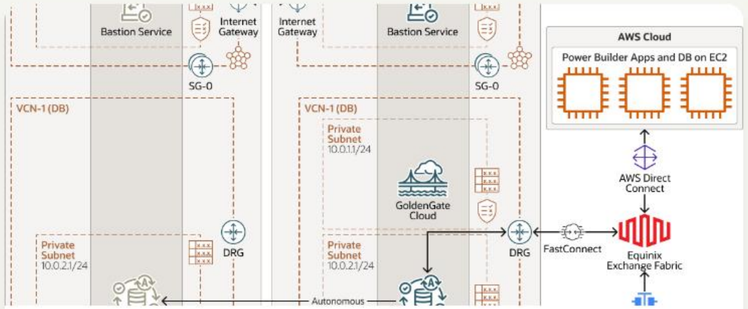
OCI Cloud

Next-gen cloud infrastructure with low-latency networking.



Exadata

Optimized hardware designed specifically for database performance.



Multi-Cloud

Strategic partnerships with Microsoft Azure and AWS.

DEMO APP: SMART AI SEARCH

1. User Query

Natural language input (e.g., "Sad space story")

2. Embedding

Convert text to 1024-dim Vector using Python

3. Oracle 23ai

Semantic search using native Vector operators

4. RAG Result

Retrieve semantically similar book records

MODERN PYTHON & SQL INTEGRATION

```python
# AI Semantic Search with Oracle-Python SDK

import oracledb

sql = """SELECT title FROM library
         ORDER BY vector_distance(embedding, :query_vec)
         FETCH FIRST 1 ROWS ONLY"""

with connection.cursor() as cursor:

    cursor.execute(sql, [my_vector])

    result = cursor.fetchone()
```

ENTERPRISE USE CASES

Industry	Large Scale Challenge	Oracle Solution
Banking	Millions of Tx per second	Real Application Clusters (RAC)
Retail	Unstructured JSON Catalogs	JSON-Relational Duality
Healthcare	Semantic Patient History Search	AI Vector Search + RAG
Tech	Global Availability	OAD Distributed Services

Questions?

Modern Oracle: Reliability Meets Innovation

Baris HANSU

IMAGE SOURCES

https://d1pnnwteuly8z3.cloudfront.net/images/ae7f7c8d-97db-47a0-b325-f3fd4c2fdbf7/ca87e117-c176-4d59-aa68-ae9b5819b66d.png

Source: www.draft1.ai

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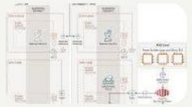

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